

or flip-chip packaging technology enhances lumens output as well as product reliability. CSP eliminates the traditional sub-mount by attaching the LED die directly to the PCB, allowing overall system cost reductions.

Moving forward

According to survey participants, the following challenges could derail growth of the LED lighting industry in India:

1. Sub-standard products from unauthorised manufacturing units and low-cost, poor-quality Chinese imports shaking consumer confidence

2. Lack of consumer awareness about the LED efficiency with respect to lux, wattage, life expectancy, etc. As a result, consumers end up buying products with higher wattage but lower efficiency

3. The inability to make LED

chips and micro-chips resulting in their import at high cost. This also limits the development of a wider variety of LED light fixtures, reducing the number of colours used, apart from hampering performance and crippling innovations in LED lighting

4. Absence of a sufficient number of LED packaging units resulting in a high dependence on imports

5. Use of inefficient drivers resulting in higher energy usage as well as product failure

6. Acute shortage of LED chip manufacturing experts from fields such as chemical, electronics, optics, lighting and thermal management

Successfully resolving the aforementioned issues through appropriate industry initiatives and government interventions will help the LED lighting industry achieve sustainable growth, thus enabling the country to move towards greater self-sufficiency in power generation. This will make more power available to the public. **EFY**

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